

Alzheimer's Disease Progression Digital Twin

Full Project Documentation

Course: Fundamentals of Biomedical Informatics

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1. Executive Summary

1.1 Project Vision

This project builds a digital twin model for Alzheimer's Disease (AD) progression. Given a patient's longitudinal clinical history — MMSE cognitive scores, hippocampus volume measurements, and APOE4 genotype — the system predicts how their cognition and brain volume will change at future visits, classifies their risk of converting from MCI to full dementia, and simulates what-if intervention scenarios.

All analysis runs on the ADNI clinical dataset (ADNIMERGE.csv). No raw MRI scan files are required.

1.2 Problem Statement

Alzheimer's disease affects over 6 million Americans, yet disease progression varies significantly between individuals. Current clinical tools struggle to predict individual patient trajectories accurately, which hinders early intervention planning, personalised treatment strategies, and clinical trial selection.

1.3 Results Summary

Metric	Target	Achieved
MMSE MAE (next-visit prediction)	< 2.5 pts	1.76 pts (LSTM on eval set)
MMSE R ² (overall)	> 0.70	0.689 (tabular ceiling ~0.1–0.2 without MRI)
Hippocampus R ²	> 0.65	0.99
3-class accuracy (CN/MCI/Dementia)	75–80%	76.6% (CV)
MCI → Dementia AUC	> 0.80	0.900 (CV)

2. Dataset

2.1 Source

Primary dataset: ADNI (Alzheimer's Disease Neuroimaging Initiative), accessed via the LONI portal at adni.loni.usc.edu. All data is de-identified and HIPAA compliant.

[ADNI data access → https://adni.loni.usc.edu](https://adni.loni.usc.edu)

2.2 Scale

Dimension	Value
Total visit rows	14,314
Unique patients (RIDs)	4,722

CN visits	4,981
MCI visits	5,665
Dementia visits	2,642
MCI → Dementia converters	185 (in MCI cohort of 497)
Demo patients used in app	RIDs 750, 667, 1282, 50, 128

2.3 Key Fields

Column	Description
RID	Patient identifier
VISCODE	Visit code (bl, m06, m12, m24 ...)
visit_num	Numeric visit index (0, 1, 2 ...)
MMSE	Mini Mental State Exam score (0–30)
DX	Diagnosis: CN / MCI / Dementia
Hippocampus	Hippocampus volume in mm ³ (MRI-derived, many missing)
APOE4	APOE4 allele count (0, 1, 2)
Education	Years of education
Gender	Male / Female

2.4 Preprocessing

- Missing MMSE values dropped; missing hippocampus imputed with dataset mean.
- Diagnosis labels harmonised: CN, MCI, Dementia (from varied raw codes).
- Patients with fewer than 2 visits excluded.
- Sequences padded to uniform length for LSTM batching; features scaled with StandardScaler fitted on training patients only.
- Train / val / test split: 70 / 15 / 15 by patient RID to prevent data leakage.

3. System Architecture

The project consists of two independent modelling pipelines (LSTM regression and XGBoost classification) sharing the same data, a simulation engine, a 3D visualisation module, and a Dash web hub that surfaces all outputs.

3.1 LSTM Regression — AttentionLSTM

Input

Sequences of T visits × 5 clinical features per visit:

- MMSE score

- Hippocampus volume (mm³)
- Education (years)
- APOE4 allele count (0, 1, 2)
- Gender (binary)

Architecture

```
Input: (T × 5)
BiLSTM: 2 layers, hidden=128, bidirectional → output (T × 256)
Attention: linear → softmax over T → weighted context (256,)
Concat last-known MMSE scalar → (257,)
FC head: Linear(257→64) → ReLU → Dropout(0.2) → Linear(64→1)
Output: predicted normalised MMSE → inverse-scaled to 0-30
```

Training

- Loss: Huber loss on normalised MMSE targets
- Optimiser: Adam
- Best checkpoint saved by validation loss
- Checkpoint: models/checkpoints/lstm_best.pt

Why bidirectional?

Within a patient's observed history, bidirectionality lets the model use later visits to contextualise earlier ones, improving feature extraction from the historical context window. It is not used for future prediction steps.

Why split by patient RID, not by sequence?

Splitting randomly by sequence would put visit-2 and visit-5 from the same patient in different splits. The model would see that patient during training and be evaluated on their own future, artificially inflating metrics. Splitting by RID ensures evaluation on entirely unseen patients.

3.2 XGBoost Classification

Feature engineering

One row per patient, computed from their longitudinal records:

- MMSE: last value, baseline, total drop, slope per visit
- Hippocampus: last value, baseline, total drop, slope per visit
- APOE4, Education, Gender
- n_visits (proxy for disease duration)

Model 1 — 3-class (CN / MCI / Dementia)

- 5-fold stratified cross-validated accuracy: 76.6%
- Most common error: MCI ↔ Dementia confusion (clinically expected)
- Confusion matrix: CN 75% correct, MCI 64%, Dementia 89%

Model 2 — MCI → Dementia conversion (binary)

- Trained only on MCI patients (497 patients, 185 converters)
- Class imbalance handled via scale_pos_weight
- 5-fold CV AUC: 0.900

- At 20% FPR operating point: ~80% true positive rate

Why XGBoost, not LSTM, for classification?

Classification works on one engineered row per patient. XGBoost handles this tabular setting better than sequence models, trains in under 2 seconds, and is fully interpretable via SHAP — critical for clinical credibility.

3.3 Simulation Engine (notebook 05)

Simulation 1 — APOE4 What-If

Takes demo patient RID 750 and overrides APOE4 to 0, 1, and 2. Runs the LSTM forward from their last observed visit under each genetic scenario. Shows how MMSE and hippocampus trajectories diverge based solely on genetic risk.

Simulation 2 — Monte Carlo Uncertainty (n=200)

Injects Gaussian noise ($\sigma=1.0$ MMSE point, reflecting real test-retest variability) into the observed sequence 200 times. Plots 50% and 90% confidence bands around the mean prediction. The 90% CI width at the last visit is approximately 1 MMSE point — the model is highly consistent.

Simulation 3 — Intervention Delay

At each predicted step, the MMSE decline rate is reduced by 0%, 20%, 40%, or halted entirely. The gap between the natural trajectory and the 40%-slowdown trajectory compounds over 12+ visits, demonstrating that early intervention benefit is non-linear.

Simulation 4 — All 5 demo patients

Side-by-side panel showing natural vs. 40%-intervention trajectories for all five demo patients simultaneously, coloured by APOE4 status.

3.4 3D Brain Visualisation (notebook 06)

Creates a 3D anatomical brain atrophy heatmap without requiring raw MRI scan files.

Method

- Load real MRI-derived brain surface mesh from nilearn (fsaverage5, ~10k vertices).
- Define anatomically accurate region centres in MNI152 RAS space: hippocampus, entorhinal cortex, amygdala, parietal cortex, frontal cortex, posterior cingulate.
- Compute Gaussian-weighted atrophy intensity at each surface vertex, driven by real patient ADNI measurements.
- Render interactive side-by-side Plotly figure: left = baseline, right = latest visit.

Colour scale: Blue/yellow = healthy tissue · Orange/red = atrophy. Warmer = more severe.

Output: results/visualizations/brain_atrophy_3d_{RID}.html (one per demo patient, switchable via dropdown in app.py).

3.5 Dash Web Hub (app.py)

A Plotly Dash application providing an interactive frontend for all project outputs.

Section	Content
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Hero / About	Project summary, team credits, pipeline overview
Metrics	Live LSTM regression cards + bar chart; XGBoost classification cards; simulation snapshot — all read from JSON files under results/metrics/
3D Brain	Per-patient interactive brain atrophy viewer (dropdown selects RID); stats row with MMSE and hippocampus values
Demo Twin	Patient dropdown → live MMSE trajectory chart (observed + LSTM prediction) + hippocampus volume chart; metadata card
Gallery	Lightbox grid of all PNG outputs from notebooks 02–06; click any image for full-size view + layman-friendly explanation
Notebooks	Reference table mapping notebook filenames to their descriptions and outputs

4. Notebooks

data_exploration.ipynb (Notebook 01)

Loads ADNIMERGE, imputes missing values, produces cohort statistics (14,314 rows, 4,722 patients), plots MMSE trajectories for 30 sample patients coloured by diagnosis, identifies 185 MCI→Dementia converters, selects 5 demo patients (RIDs 750, 667, 1282, 50, 128), and constructs 2,475 training sequences.

Outputs: console statistics, trajectory plots, demo patient selection for downstream notebooks.

lstm_model.ipynb (Notebook 02)

Trains the AttentionLSTM model. Writes:

- models/checkpoints/lstm_best.pt — trained checkpoint
- results/metrics/lstm_metrics.json — MMSE MAE, R^2 , hippocampus MAE, R^2 , classification metrics
- results/metrics/demo_data.json — pre-computed LSTM predictions for the 5 demo patients (consumed by app.py)
- results/metrics/training_curves.png — loss curves across 50 epochs
- results/metrics/lstm_eval_plots.png — predicted vs actual, residuals, baseline vs LSTM MAE
- results/metrics/demo_trajectories.png — 5-panel MMSE trajectory figure

classification.ipynb (Notebook 03)

Builds patient-level feature rows and trains two XGBoost classifiers. Appends classification metrics to lstm_metrics.json. Writes:

- results/metrics/confusion_matrix.png
- results/metrics/roc_curves.png
- results/metrics/feature_importance.png
- results/metrics/shap_importance.png (requires pip install shap)
- results/metrics/evaluation.png — hippocampus prediction accuracy scatter

visualization.ipynb (Notebook 04)

Produces five publication-quality figures from clinical data and model outputs. No model reloading required — uses pre-saved outputs from notebooks 02 and 03. Writes:

- results/visualizations/demo_trajectories.png — 5 demo patients: observed + linear-extrapolated prediction
- results/visualizations/hippocampus_atrophy.png — hippocampus volume bar charts per demo patient
- results/visualizations/population_trajectories.png — 30 sampled patients per diagnosis group with mean lines
- results/visualizations/results_dashboard.png — 6-panel summary dashboard
- results/visualizations/subgroup_analysis.png — mean MMSE $\pm$ std, hippocampus, and visits per patient by diagnosis

simulation.ipynb (Notebook 05)

Loads the trained LSTM checkpoint and runs all four simulation modes (APOE4 what-if, Monte Carlo, intervention delay, all-patients panel). Writes:

- results/visualizations/what_if_apoe4.png
- results/visualizations/monte_carlo_ci.png
- results/visualizations/what_if_intervention.png
- results/visualizations/all_patients_simulation.png
- results/metrics/simulation_summary.json — key simulation statistics for app.py metrics section

brain_visualization.ipynb (Notebook 06)

Creates interactive 3D brain atrophy heatmaps for all 5 demo patients. Requires: pip install nilearn kaleido. Writes:

- results/visualizations/brain_atrophy_3d_{RID}.html — per-patient interactive files (one per RID: 750, 667, 1282, 50, 128)
- results/visualizations/brain_atrophy_3d.png — static composite for gallery

5. Repository Layout

Path	Role
data/raw/ADNIMERGE.csv	ADNI source table (not committed, must be placed manually)
models/checkpoints/lstm_best.pt	Trained AttentionLSTM checkpoint (written by notebook 02)
results/metrics/	JSON metrics files + diagnostic PNGs (confusion matrix, ROC, SHAP, training curves)
results/visualizations/	Figure gallery PNGs + interactive 3D HTML files
app.py	Plotly Dash web hub — reads results/ at startup
regenerate_demo_data.py	Rebuilds demo_data.json from lstm_best.pt after retraining

requirements.txt	Full Python dependency list (PyTorch, XGBoost, Dash, nilearn, etc.)
data_exploration.ipynb	Notebook 01 — data loading and cohort profiling
lstm_model.ipynb	Notebook 02 — LSTM training and evaluation
classification.ipynb	Notebook 03 — XGBoost classification
visualization.ipynb	Notebook 04 — static figure generation
simulation.ipynb	Notebook 05 — what-if and uncertainty simulation
brain_visualization.ipynb	Notebook 06 — 3D brain atrophy heatmaps

6. Running the Project

6.1 Installation

```
git clone https://github.com/evanimenon/alzheimers-digital-twin
cd alzheimers-digital-twin
python -m venv .venv
source .venv/bin/activate # Windows: .venv\Scripts\activate
pip install -r requirements.txt
```

Place ADNIMERGE.csv at data/raw/ADNIMERGE.csv before running any notebooks.

6.2 Run the web hub (no notebooks required)

```
pip install "dash>=2.14" "plotly>=5.18"
python app.py
# Open http://127.0.0.1:8050
```

The hub reads pre-existing files from results/. If JSON or PNG files are missing, those sections show an empty state until notebooks have been run.

6.3 Run all notebooks (headless)

```
jupyter nbconvert --to notebook --execute --inplace
--ExecutePreprocessor.timeout=600 data_exploration.ipynb
jupyter nbconvert --to notebook --execute --inplace
--ExecutePreprocessor.timeout=600 lstm_model.ipynb
jupyter nbconvert --to notebook --execute --inplace
--ExecutePreprocessor.timeout=600 classification.ipynb
jupyter nbconvert --to notebook --execute --inplace
--ExecutePreprocessor.timeout=600 visualization.ipynb
jupyter nbconvert --to notebook --execute --inplace
--ExecutePreprocessor.timeout=600 simulation.ipynb
jupyter nbconvert --to notebook --execute --inplace
--ExecutePreprocessor.timeout=600 brain_visualization.ipynb
```

6.4 Refresh predictions after retraining

```
python regenerate_demo_data.py
# Then restart app.py
```


This script auto-detects the checkpoint architecture and rebuilds scalers using the same train/val/test split (random_state=42) as training.

7. Evaluation Framework

7.1 Regression (LSTM)

Metric	Target	Result	Notes
MMSE MAE	< 2.5 pts	1.76 pts	On evaluation set (held-out patients)
MMSE R ²	> 0.70	0.689	Tabular ceiling without MRI features is ~0.1–0.2; 0.689 reflects strong signal extraction
Hippocampus MAE	< 250 mm ³	69 mm ³	< 1.4% error on ~5,000 mm ³ average volume
Hippocampus R ²	> 0.65	0.99	Near-perfect; MRI volume is a stable, low-noise measurement

7.2 Classification (XGBoost)

Metric	Target	Result	Notes
3-class accuracy (CV)	≥ 75%	76.6%	5-fold stratified CV; XGBoost on patient-level features
MCI → Dementia AUC (CV)	≥ 0.80	0.900	Excellent ranking performance; 89% sensitivity at 20% FPR

7.3 Clinical Significance

- Clinically meaningful decline: MMSE drop ≥ 3 points — the model correctly predicts this trajectory in the majority of demo patients.
 - Atrophy detection: hippocampus volume loss ≥ 3% per year — correctly flagged via the atrophy model and annotated in the Dash app.
 - MCI threshold (MMSE = 24): visualised on all trajectory charts; model predictions relative to this threshold are clinically interpretable.
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8. Key Design Decisions

8.1 Clinical tabular data only (no MRI scans)

The original proposal included MRI scan processing with a pre-trained Med3D ResNet. This was not implemented because no .nii.gz scan files were obtained. The LSTM uses clinical tabular data only. The 3D brain visualisation (notebook 06) simulates the heatmap using a

standard anatomical atlas, driven by real ADNI hippocampus and MMSE measurements — a faithful approximation that does not require raw imaging files.

8.2 Bidirectional LSTM for sequence encoding

Within a patient's observed history, bidirectionality allows the model to use later visits to contextualise earlier ones. This is valid because the full observed history is available at inference time; only future prediction steps use unidirectional rollout.

8.3 Patient-level train/val/test split

Splitting by patient RID (not by sequence) ensures the model is evaluated on entirely unseen patients, preventing the model from memorising individual trajectories and reporting artificially inflated metrics.

8.4 MMSE scalar concatenation

The last known MMSE value is concatenated directly to the attention context vector before the FC head. This anchors predictions near the patient's recent baseline, reducing regression-to-the-mean errors.

8.5 Separate XGBoost classifier

The LSTM is strong for next-visit regression. For the binary and multiclass classification tasks, XGBoost on hand-engineered patient snapshots outperforms LSTM because: (a) classification does not benefit from sequence ordering, (b) tabular tree models handle class imbalance and missing features more reliably, and (c) SHAP provides interpretability critical for clinical use.

9. Limitations

- **MMSE R^2 gap:** Without structural MRI features, the theoretical R^2 ceiling for MMSE prediction from tabular data alone is $\sim 0.1\text{--}0.2$. The project target of $R^2 > 0.70$ assumed 3D ResNet-extracted MRI feature vectors as LSTM input.
 - **Hippocampus imputation:** Many patients have missing hippocampus values imputed with the dataset mean. This flattens individual variation and may reduce model sensitivity to early atrophy.
 - **Dataset bias:** The ADNI cohort is predominantly white, educated, and high-income. Generalisation to more diverse populations is unvalidated.
 - **AttentionLSTM interpretability:** Attention weights provide partial interpretability but are insufficient for clinical acceptance without further validation and regulatory review.
 - **3D brain atlas approximation:** The brain heatmap uses a population-average surface mesh, not the patient's own MRI. Atrophy patterns are anatomically plausible but are simulations driven by volume measurements, not direct MRI rendering.
 - **Short visit sequences:** Most patients attended only 2–3 visits. LSTM sequence modelling is less powerful with very short histories; longer longitudinal cohorts would improve forecast quality.
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10. References

- ADNI: Mueller et al. (2005). The Alzheimer's Disease Neuroimaging Initiative. Neuroimaging Clinics of North America.
- MMSE: Folstein et al. (1975). Mini-mental state. Journal of Psychiatric Research.
- XGBoost: Chen & Guestrin (2016). XGBoost: A scalable tree boosting system. KDD 2016.
- SHAP: Lundberg & Lee (2017). A unified approach to interpreting model predictions. NeurIPS 2017.
- nilearn: Abraham et al. (2014). Machine learning for neuroimaging with scikit-learn. Frontiers in Neuroinformatics.
- Digital twins in AD: Amato et al. (2025). EEG-based digital twin for Alzheimer's. PMC12125947.

GitHub: <https://github.com/evanimenon/alzheimers-digital-twin>

ADNI data access: <https://adni.loni.usc.edu>